Detailed Note on 5-Fold CV Rank Selection for the Matrix Model with $\log_{10}(EC) - \log_{10}(EO)$

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1 Purpose

This note documents the matrix rank-selection procedure implemented in `run_hbn_trace_regression_5fold_`
The goal is to select a single matrix rank for the trace regression model when the predictor is restricted to the

$$\log_{10}(EC) - \log_{10}(EO)$$

contrast. In this document, "5-fold CV" means a single-layer cross-validation procedure:

- the data are partitioned into five folds;
- for each candidate rank, the model is fit on four folds and evaluated on the held-out fold;
- the final rank is the one with the smallest mean validation MSE across the five folds.

There is no inner loop and no nested model selection in this analysis.

2 Data input and predictor construction

The input file is

`hbn_bandpower_8band_roi10.mat`,

which contains $N = 397$ subjects. Each subject contributes:

- age as the scalar response y_i;
- an 10×8 eyes-closed (EC) ROI-by-band matrix;
- an 10×8 eyes-open (EO) ROI-by-band matrix.

The ten rows correspond to the ROI groups and the eight columns correspond to the canonical frequency bands.

Only one predictor representation is used here. Let $X_i^{EC} \in \mathbb{R}^{10\times 8}$ and $X_i^{EO} \in \mathbb{R}^{10\times 8}$ denote the raw ROI band-power matrices. The analysis first applies the entrywise transform

$$\widetilde{X}_i^{EC} = \log_{10}(X_i^{EC}), \qquad \widetilde{X}_i^{EO} = \log_{10}(X_i^{EO}),$$

with a numerical floor

$$\text{floor} = \max\{0.5 \times \min(\text{positive raw entries}),\ 10^{-12}\}$$

inserted before taking logarithms. The matrix covariate used in CV is then

$$X_i = \widetilde{X}_i^{EC} - \widetilde{X}_i^{EO}.$$

Thus the current analysis is a log-contrast analysis, not a relative-power analysis.

3 Model

For a candidate rank r, the fitted model is the no-intercept trace regression

$$y_i = \langle X_i, B \rangle + \varepsilon_i, \qquad \operatorname{rank}(B) \le r.$$

Here $B \in \mathbb{R}^{10\times 8}$ is the coefficient matrix and

$$\langle X_i, B \rangle = \operatorname{tr}(X_i^\top B).$$

4 5-fold splitting rule

The five folds are constructed by a release-stratified split with random seed 20260320. The release label is used only for fold construction. For each release,

- the subject indices are shuffled;
- the shuffled indices are partitioned into five nearly equal pieces;
- the k-th piece from every release is concatenated to form validation fold k.

This preserves release composition across folds up to the discrete partitioning constraint.

5 Centering within each fold

Because the model does not include an intercept, centering is carried out inside each training fold. For fold k, let $\mathcal{I}_k^{tr}$ and $\mathcal{I}_k^{val}$ denote the training and validation indices. Define

$$\bar{X}_k = \frac{1}{|\mathcal{I}_k^{tr}|} \sum_{i \in \mathcal{I}_k^{tr}} X_i, \qquad \bar{y}_k = \frac{1}{|\mathcal{I}_k^{tr}|} \sum_{i \in \mathcal{I}_k^{tr}} y_i.$$

Then both training and validation data are centered using the *training-fold* means:

$$X_i^{(k,c)} = X_i - \bar{X}_k, \qquad y_i^{(k,c)} = y_i - \bar{y}_k.$$

The model is fit on the centered training data, and prediction on the validation fold is performed using the centered validation matrices $X_i^{(k,c)}$. Hence the CV comparison is consistent with a no-intercept formulation.

6 Fixed-rank fitting algorithm

Given a rank r and centered training data, the solver minimizes

$$\min_{\operatorname{rank}(B)\le r} \frac{1}{2n_k} \sum_{i \in \mathcal{I}_k^{tr}} \left(y_i^{(k,c)} - \langle X_i^{(k,c)}, B \rangle \right)^2.$$

The implementation uses projected gradient descent:

1. reshape each matrix $X_i^{(k,c)}$ into a row of a design matrix Z;
2. compute a Lipschitz constant
$$L = \sigma_{\max}(Z)^2 / n_k;$$
3. set the step size $\eta = 1/L$;

4. update
$$\widetilde{B} = B - \eta \nabla f(B);$$
5. project $\widetilde{B}$ onto the set $\{B : \operatorname{rank}(B) \le r\}$ by truncated SVD;
6. stop when the relative coefficient change or relative objective change drops below 10^{-7}, or when 4000 iterations are reached.

The candidate ranks searched in this run are

$$r \in \{1, 2, \ldots, 8\},$$

because the matrix size is 10×8.

7 Validation criterion and rank choice

For each rank r, fold k, and fitted matrix $\widehat{B}_{r,k}$, the validation MSE is

$$\mathrm{MSE}_{r,k} = \frac{1}{|\mathcal{I}_k^{val}|} \sum_{i \in \mathcal{I}_k^{val}} \left(y_i^{(k,c)} - \langle X_i^{(k,c)}, \widehat{B}_{r,k} \rangle\right)^2.$$

The summary criterion is the mean validation MSE

$$\overline{\mathrm{MSE}}_r = \frac{1}{5} \sum_{k=1}^{5} \mathrm{MSE}_{r,k}.$$

The selected rank is

$$\widehat{r} = \arg \min_{r \in \{1,\ldots,8\}} \overline{\mathrm{MSE}}_r.$$

If two ranks have exactly the same mean validation MSE, the smaller rank is preferred because the summary table is sorted by `(mean_val_mse, rank)`.

8 Result of the current run

The current run produced the following summary:

- selected rank: 1;
- mean CV MSE at rank 1: 7.590061;
- standard deviation of foldwise validation MSE at rank 1: 0.877828;
- mean CV R^2 at rank 1: 0.267672.

The foldwise validation MSEs at the selected rank are

$$[6.703889,\ 6.650621,\ 8.094055,\ 8.629773,\ 7.871965].$$

The foldwise validation R^2 values are

$$[0.330486,\ 0.318940,\ 0.182993,\ 0.182682,\ 0.323259].$$

The rank summary over all candidates is given in Table 1.

Table 1: 5-fold CV summary for the matrix log-contrast analysis.

Rank	Mean val MSE	SD val MSE	Mean val R^2	Mean iterations
1	7.5901	0.8778	0.2677	1635.8
2	8.5531	0.8582	0.1756	3978.8
4	9.1332	0.9389	0.1207	4000.0
3	9.2244	1.0183	0.1125	4000.0
7	9.2612	0.9979	0.1064	4000.0
8	9.3533	1.0239	0.0980	4000.0
5	9.3893	1.0397	0.0956	4000.0
6	9.4784	1.0362	0.0866	4000.0

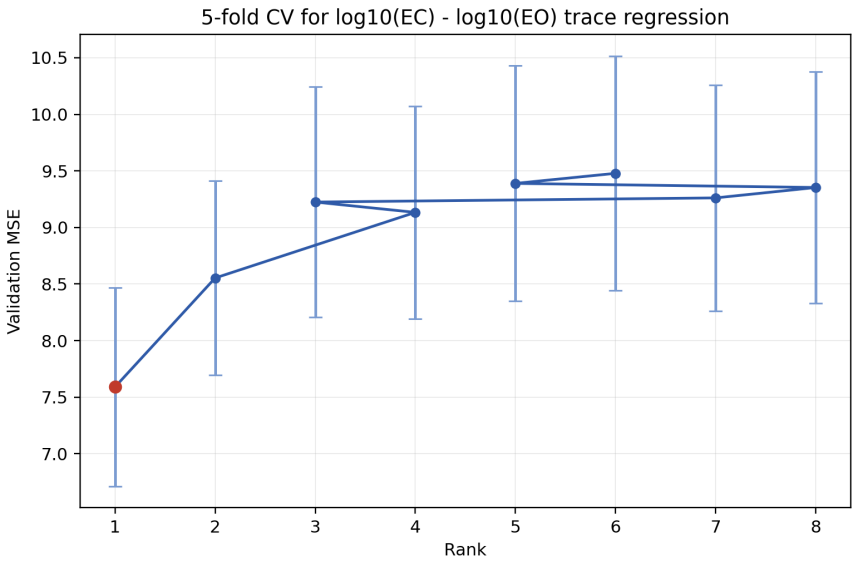

Figure 1: Mean validation MSE by rank for the 5-fold CV matrix log-contrast analysis.

Remark. In Table 1, an average iteration count of 4000.0 means that the corresponding rank hit the iteration cap in the formal run with `max_iter=4000`; it should not be interpreted as a proof that the optimizer would never converge under a larger cap. Additional diagnostic reruns with larger iteration budgets show that higher ranks can indeed converge, but their validation MSE values remain larger than the rank-1 MSE reported above, so the rank-1 conclusion is unchanged.

Figure 1 plots the mean validation MSE by rank using the output generated by the script.

9 Post-selection refit

After rank selection, the script recenters the full data using the full-sample means and refits the rank-1 model on all subjects. This step produces

- the selected coefficient matrix $\widehat{B}$;
- the full-sample centered fitted values;
- the coefficient CSV file for downstream interpretation.

In the current run, the full-sample centered training MSE at rank 1 is 6.965971, and the corresponding centered R^2 is 0.326381.

10 Output files

The script writes the following files:

- `cv5_fold_manifest.csv`: subject-to-fold assignment;
- `cv5_rank_details.csv`: foldwise validation metrics for every candidate rank;
- `cv5_rank_summary.csv`: rank-level summary sorted by mean validation MSE;
- `selected_rank_1_coefficients.csv`: fitted coefficient matrix at the selected rank;
- `cv5_selected_model.mat`: MATLAB export of the selected model and CV summaries;
- `cv5_summary.txt`: plain-text summary of the current run;
- `cv5_mean_mse_by_rank.png`: CV curve used in Figure 1.

11 Interpretation

Under this single-layer 5-fold CV criterion, the best-performing matrix rank is 1. Hence, when the predictor is restricted to $\log_{10}(EC) - \log_{10}(EO)$, the data favor the simplest low-rank representation among the searched ranks. In other words, the validation-error criterion does not support adding extra matrix rank complexity beyond one dominant contrast direction in this setup.